A Kitchen Sponge

01 What is inside all those holes?



Tiny holes lead into connected spaces.

Look closer. The magnifier makes the sponge's open spaces easier to notice.

A little explanation. A kitchen sponge has pores: tiny holes leading into connected spaces. Water can travel through these spaces, like little tunnels. Most kitchen sponges are made by people from cellulose, a plant fibre, or foam. Many sea sponges are simple animals; this kitchen sponge is a made object.

Try this. Find two holes, then imagine a path between spaces inside.

02 How does water move into the sponge?



Water clings to the walls and creeps into narrow gaps.

Look closer. The sponge's wet edge grows darker as it touches water.

A little explanation. Water clings to the sponge's walls and creeps along narrow gaps. We call this spreading wicking. To absorb means to take water in. Some water is held inside the connected spaces, so you cannot see all of it from outside.

Try this. With Big leading, use a new, clean sponge, a tray, two clear cups, and a measuring spoon. Dip, lift, and squeeze over the empty cup. Wipe up drips so the floor stays dry.

03 Why does a squeeze bring the water back?



A squeeze makes the spaces smaller and pushes water out.

Look closer. Drops fall from the squeezed sponge into the cup.

A little explanation. Squeezing presses the sponge's walls closer together. There is less room inside, so water is pushed out. Let go, and the springy material opens back up. It can soak up water again.

Try this. Compare the sponge's shape during a squeeze and after you let go.

04 Would a damp sponge soak up a spill faster?



Same-size pieces, the same water, and the same time make a fair test.

Look closer. Two same-size pieces meet equal little spills for the same time.

A little explanation. A damp sponge often starts soaking faster than a bone-dry one. Damp means a little wet, not full of water. Compare fairly: use the same kind and size of sponge, equal amounts of water, and the same time. Look at how much of each spill is left.

Try this. Predict which spill will shrink faster, then compare what is left.